

KASHAF: A Hierarchical Pipeline for Arabic Interpretable Semantic Textual Similarity

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Abstract

We introduce KASHAF, to our knowledge the first interpretable Semantic Textual Similarity (iSTS) resource and end-to-end system for Arabic: a corpus of 7,811 human-annotated sentence pairs (inter-annotator $\kappa=0.83$ on a doubly-annotated subset) under a three-level hierarchical schema (sentence, phrase, and word), together with a modular pipeline that produces this layered analysis plus a composite similarity score, the KASHAF Hierarchical Similarity Score (KHSS), calibrated to the ordinal SEM scale. On a held-out cross-batch test split (1,002 pairs, disjoint from training and dev), KASHAF reaches Macro F1 = 0.659 (± 0.016 , 5 seeds) and Spearman $\rho=0.6715$; a rule-based antonym/negation polarity gate provides an empirical score-capping rule for high-similarity contradictions, and the MARBERTv2 backbone leads three alternative Arabic encoders. As an off-the-shelf diagnostic rather than a fine-tuned comparison, four contemporary LLMs under matched prompting reach at most macro-F1 0.215 (standard prompting) and 0.336 (chain-of-thought), a 32–44-point gap that reflects systematic over-prediction of SEM-PARA at the expense of the dominant SEM-PARTIAL class. We release the schema, guidelines, code, models, and a reproducibility pipeline; the corpus is available from KSAA on request.

1 Introduction

Standard Semantic Textual Similarity (STS) produces a single score that tells the user *how* similar two sentences are, but not *why*. Interpretable STS (iSTS) was introduced at SemEval-2015 (Agirre et al., 2015, 2016) to address this gap: each sentence pair is segmented into chunks, chunks are aligned across the pair, and each alignment is labeled with a relation type. The resulting three-level explanation supports downstream applications including educational feedback, MT evalua-

tion, paraphrase mining, and legal/medical document comparison.

Despite a decade of iSTS work for English, Spanish, and a handful of other languages, **no published iSTS schema, corpus, or system exists for Arabic to our knowledge**. Arabic exhibits rich morphology, a non-concatenative templatic word formation system, free word order with strong information-structural preferences, and pervasive null subjects, all of which complicate alignment and tagging at fine granularity.

In this paper we introduce KASHAF (Arabic: كَشَاف, *revealer*), to our knowledge the first iSTS resource and end-to-end system for Arabic. On a held-out cross-batch test set (1,002 pairs from two batches the system never saw during training or hyperparameter tuning), KASHAF achieves macro-F1 = 0.659 and Spearman $\rho = 0.6715$ for four-way SEM classification. As an off-the-shelf reference on the same split, four contemporary LLMs (GPT-4o, GPT-4o-mini, Claude Sonnet 4.5, Claude Haiku 4.5) under matched prompting reach at most macro-F1 = 0.215 (3-shot) and 0.336 (chain-of-thought), a 32–44-point gap, not a fine-tuned comparison, that reflects systematic over-prediction of SEM-PARA at the expense of the dominant SEM-PARTIAL class (71.7% of test data). This over-prediction is most severe under zero-shot and 3-shot prompting and only partially alleviated by chain-of-thought.

Contributions. (1) A hierarchical Arabic iSTS schema (4 SEM, 6 PHR, 7 WRD tags) with an explicit anchor-and-mobile alignment protocol, applied to 7,811 annotated pairs with $\kappa=0.83$ on SEM. (2) A modular pipeline combining MARBERTv2 (Abdul-Mageed et al., 2021) embeddings, LightGBM tag classifiers, a CRF-based BIO segmenter, and a polarity-aware calibration gate built on Arabic WordNet (Black et al., 2006) with morphological expansion over 13,329

AWN seed lemmas. (3) The KHSS composite score combining hierarchical evidence under an independent polarity check; we transparently analyse that the phrase/word terms add interpretation rather than predictive lift, and that the polarity gate’s aggregate Spearman effect is small and not statistically significant on the test split ($\Delta\rho=+0.0231$, 95% CI $[-0.007, +0.053]$). (4) Controlled paired ablations showing (a) the systematic LLM SEM-PARTIAL $\rightarrow$ SEM-PARA collapse holds across all four models, and (b) we find *no significant evidence* that Stage 2 contrastive pre-training improves phrase or SEM classification ($p \in \{0.40, 0.80, 0.89\}$ on three core metrics, 3-seed paired t -test; non-significance under limited seeds is not proof of equivalence).

We release annotation guidelines (English and Arabic), source code, trained models, and a corpus sample under a non-commercial research license; the full corpus is available to researchers on request from the King Salman Global Academy for the Arabic Language (KSAA; Appendix K).

2 Related Work

Interpretable STS. The iSTS task was introduced by Agirre et al. (2015) and extended at SemEval-2016 (Agirre et al., 2016). Lopez-Gazpio et al. (2017) consolidated the framework, adding a verbalization module and two user studies showing that interpretable alignments help humans perform downstream judgments, establishing usability, not just intrinsic accuracy, as the goal of iSTS. Participating systems combined handcrafted chunk alignment with similarity features, and subsequent work pairs chunk alignment with classification and regression (Majumder et al., 2021). None target Arabic.

Arabic STS. Cer et al. (2017) included cross-lingual Arabic-English STS, but no published Arabic-only sentence-level iSTS schema or corpus exists to our knowledge. Abdul-Mageed et al. (2021) introduced MARBERTv2, a strong Arabic semantic encoder we build on; their work targets representation quality rather than fine-grained iSTS. For sentence-level similarity specifically, AraBERT (Antoun et al., 2020) established strong Modern Standard Arabic baselines, and more recent sentence-embedding models such as the Arabic Triplet Matryoshka encoders trained under the GATE framework (Nacar et al., 2025) report competitive STS scores; we include these, with

CAMeLBER (Inoue et al., 2021), as encoder baselines (Section 6). None, however, produces the interpretable phrase- and word-level trace that is our focus.

Word and phrase alignment. SimAlign (Jalili Sabet et al., 2020) produces word alignments from contextualized embeddings without parallel training data. AwesomeAlign (Dou and Neubig, 2021) adds light supervision. For morphologically rich languages, classical statistical aligners underperform where semantic precision matters. KASHAF builds on SimAlign and adds a discriminative LightGBM classifier.

LLMs for fine-grained similarity. Recent work on LLMs for STS reports strong coarse-benchmark results but behaviour that varies by setting: generative models can lead on knowledge-grounded similarity (Gatto et al., 2023), while fine-grained or linguistically conditioned variants remain harder (Tu et al., 2024). For Arabic, strong fine-tuned encoders such as MARBERTv2 (Abdul-Mageed et al., 2021) remain competitive with or superior to general-purpose LLMs on fine-grained understanding tasks; our work quantifies this gap for interpretable STS with a breakdown of LLM failure modes.

Arabic lexical resources. Arabic WordNet (Black et al., 2006) provides a lexical-semantic backbone; we augment it with morphological expansion (~ 30 surface forms per lemma over 13,329 seed lemmas) for antonym lookup at scale.

3 Schema and Dataset

KASHAF adopts a three-level hierarchical schema (Table 1) designed to give an interpretable trace of *why* two sentences are similar. A single SEM tag captures the overall semantic relation; phrase-level alignments locate *where* agreement and divergence occur; word-level tags surface the specific lexical and morphological alternations.

Schema tags. The four SEM tags, SEM-IDENT (identical, surface variation allowed), SEM-PARA (paraphrase), SEM-PARTIAL (partial overlap), and SEM-DIFF (different meaning or contradiction), map onto a 0–3 ordinal scale, exploited by our ordinal SEM head (Section 4). PHR has six tags: MATCH, MOD (same meaning, different surface), SYN, REORDER, ADD_IN_B, DEL_IN_B. WRD has five relation types: MATCH, SYN, MORPH, FUNC, ANT,

Level	Tag	Description
Sentence (SEM)	SEM-IDENT	Identical meaning (surface variation OK)
	SEM-PARA	Paraphrase: same meaning, different wording
	SEM-PARTIAL	Partial similarity: extra/missing info on one side
	SEM-DIFF	Different meaning or contradiction
Phrase (PHR)	PHR-MATCH	Lexical overlap, same content
	PHR-MOD	Same meaning, different surface form (morph/agreement)
	PHR-SYN	Synonym-substitution
	REORDER	Same content, different position in sentence
	PHR-ADD_IN_B	Phrase present only in B
	PHR-DEL_IN_B	Phrase present only in A
Word (WRD)	WRD-MATCH	Exact same word (or trivial inflection)
	WRD-SYN	Synonym
	WRD-MORPH	Morphological inflection of same root
	WRD-FUNC	Function-word change (preposition, determiner)
	WRD-ANT	Antonym
	WRD-ADD_IN_B	Word present only in B
	WRD-DEL_IN_B	Word present only in A

Table 1: The KASHAF hierarchical iSTS schema: 4 SEM tags, 6 PHR tags, 7 WRD tags.

plus directional ADD_IN_B/DEL_IN_B for words with no counterpart (mapped to SemEval-iSTS types in Appendix D).

3.1 Annotation Methodology

Anchor-and-mobile alignment. A crucial design choice in our schema is the *asymmetric* treatment of the two sentences during alignment (Figure 1). Sentence *A* is the **anchor**: its phrase and word positions are held fixed in canonical left-to-right order. Sentence *B* is the **mobile**: each *B*-phrase is realigned to sit directly beneath its best-matching counterpart in *A*, while *B*’s *original* position is recorded. This eliminates ambiguity in defining REORDER (the tag fires when *B*’s original position differs from its anchored position), yields a stable canonical form for downstream feature engineering, and mirrors how annotators naturally read the second sentence. The same principle applies at the word level inside each phrase pair.

Annotation procedure. Trained linguists annotate each pair following a 100+-page bilingual (Arabic–English) annotation guide (AIGhamdi et al., 2026), whose SEM-tag decision rules are summarised in Appendix C: (1) assign the SEM tag; (2) segment both sentences into phrases (clausal/PP/NP units); (3) realign *B*’s phrases against *A* as anchor; (4) tag each aligned phrase pair (ADD/DEL for unaligned); (5) align words within each pair using *A*-words as anchor and tag.

Inter-annotator agreement (IAA). On a 781-pair gold subset double-annotated by two senior linguists (third resolving disagreements), we report

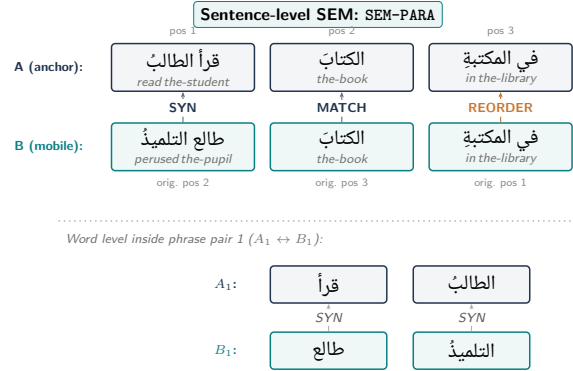


Figure 1: Three-level annotation of the SEM-PARA pair. Sentence A (anchor) is 'قرأ الطالب في المكتبة' ("the student read the book in the library") and Sentence B (mobile) is 'طالع التلميذ في المكتبة' ("in the library the pupil perused the book"). A is the fixed anchor; each B-phrase is realigned beneath its best match and its original position is recorded, yielding SYN (synonyms: *قرأ*/طالع, *الطالب*/التلميذ), MATCH, and REORDER (the fronted *في المكتبة*). The same anchor-and-mobile principle applies at the word level inside each phrase pair.

Cohen’s κ for SEM and micro-F1 for PHR/WRD tag sequences (Artstein and Poesio, 2008). Our agreement: $\kappa = 0.83$ for SEM, $F_1 = 0.79$ (PHR), $F_1 = 0.77$ (WRD). By Landis and Koch (1977) thresholds, $\kappa \in [0.81, 1.0]$ is “almost perfect”; our SEM agreement falls in this range, consistent with the iSTS literature norm of $\kappa \in [0.70, 0.85]$ (Agirre et al., 2016).

Corpus and splits. We construct 7,811 pairs from a heterogeneous mix of public Arabic sources (movie and television subtitles (Lison and Tiedemann, 2016), product reviews, encyclopaedic arti-

Split	Pairs	Phrase tags*
Train (13 batches)	5,809	~80,500
Dev (B13 + B14)	1,000	~11,200
Test (B03 + B15)	1,002	11,145
Total annotated	7,811	~102,800

Table 2: KASHAF corpus statistics. Cross-batch splits are designed so that test contains complete annotator-batches unseen during training or development; dev and test share no batches. Train uses 13 batches (B01–B02, B04–B12, B16–B17; B03 is held out for test, B13–B14 for dev, B15 for test). **Phrase tags* counts all annotated phrase-level decisions per pair, including aligned PHR-MATCH/MOD/SYN/REORDER and unilateral PHR-ADD_IN_B/PHR-DEL_IN_B; see Appendix A for per-tag breakdown.

cles (Wikipedia), and scientific texts), augmented with curated paraphrase pools and adversarial near-paraphrase distractors. *Cross-batch* evaluation: train on the remaining 13 batches, dev on B13+B14 (1,000 pairs), test on B03+B15 (1,002 pairs). SEM-PARTIAL dominates (71.7%) followed by SEM-PARA (19.8%), SEM-DIFF (4.3%), SEM-IDENT (4.3%).

Data integrity audit. We programmatically audited the splits for leakage and release the audit script with the data. **No Pair_ID is shared between train, dev, and test** (0 duplicates by exact ID match). The cross-batch protocol assigns each annotator-batch to a single split, so the partition is disjoint at the annotation-unit level by construction. Beyond identifiers, after Arabic normalisation (diacritics, alif variants, *ta marbuta*, whitespace, punctuation) we verify that no test sentence pair recurs in the training or development data, ruling out verbatim pair leakage. All main results are therefore reported on the full $n=1,002$ held-out test split.

4 The KASHAF Pipeline

KASHAF is a modular pipeline of five trained components and one rule-based component, producing both a hierarchical interpretation and a scalar KHSS similarity score (Figure 2). Only the SEM level is *symmetric*: its features use symmetric operations (cosine, Jaccard, length-difference), so sentence-level predictions are invariant to swapping A and B . The phrase and word levels are *equivariant* rather than symmetric: the BIO segmenter and aligner apply to A and B with shared weights (phrase alignment via bipartite max-weight matching), but their tags

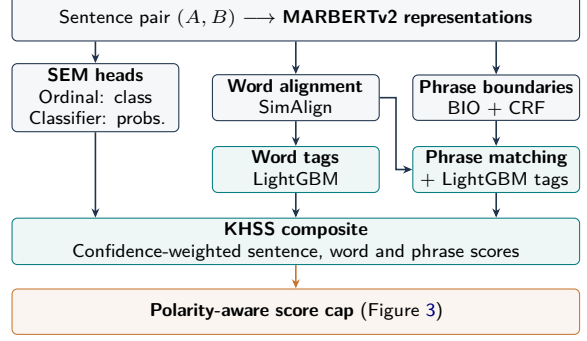


Figure 2: KASHAF overview: frozen SEM/alignment features, a fine-tuned boundary encoder, and separate class/probability heads. Polarity-gate conditions: Figure 3.

are directional by annotation convention (e.g., ADD_IN_B/DEL_IN_B mark material present only in B).

MARBERTv2 backbone. We use MARBERTv2 (Abdul-Mageed et al., 2021), a 163M-parameter encoder pre-trained on Modern Standard Arabic and Egyptian/Levantine dialectal data, both as a frozen feature extractor (ordinal SEM, word/phrase alignment) and as a fine-tuned encoder (boundary detection). We empirically select layer-8 embeddings, which we found to provide the best balance of syntactic and semantic information for Arabic word alignment in preliminary experiments on the dev set.

Stage 2 contrastive pre-training (optional). We optionally fine-tune MARBERTv2 on 2,080 SEM-IDENT + SEM-PARA pairs using MultipleNegativesRankingLoss (Henderson et al., 2017) with 4 in-batch negatives. We ablate this step in Section 6.3: contrastive pre-training yields at most a marginal, non-significant gain (≈ 0.01 macro-F1), with vanilla MARBERTv2 performing nearly identically. The headline configuration uses the Stage-2 backbone (Table 3); the encoder, not this auxiliary step, drives sentence-level performance.

Boundary detector. A BIO (Begin–Inside–Outside) sequence-labeling head combining fine-tuned MARBERTv2 with a linear classifier and a Conditional Random Field (CRF) (Lafferty et al., 2001). We concatenate per-token mean of layers 6–12, giving the best dev F1. Training: 5 epochs, batch size 32, learning rate 2×10^{-5} , Adam (Kingma and Ba, 2015).

Word alignment. For each sentence pair, we compute layer-8 token embeddings and run SimAlign (Jalili Sabet et al., 2020) (inter matching) to produce a sparse alignment $\mathcal{A}_w$. A LightGBM tag classifier scores per-alignment features: embedding cosine similarity, surface-form similarity, Arabic WordNet lookups, character n-gram Jaccard, Farasa (Abdelali et al., 2016) morphological distance, and antonym hits.

Phrase alignment and tagging. Phrase alignments are induced from word alignments via max-weight matching:

$$\text{align}(p_A, p_B) = \frac{|\mathcal{A}_w \cap (p_A \times p_B)|}{\max(|p_A|, |p_B|)}, \quad (1)$$

where $\mathcal{A}_w$ is the word-alignment set. Tag classification uses LightGBM (Ke et al., 2017) (500 boosting rounds, 31 leaves, learning rate 0.05) over phrase-pair features (full feature list in Appendix I).

Ordinal SEM head. Sentence-level SEM is naturally ordinal (SEM-IDENT $\succ$ SEM-PARA $\succ$ SEM-PARTIAL $\succ$ SEM-DIFF). We train two variants: (i) a 4-way LightGBM classifier over sentence-pair features, and (ii) an ordinal-regression variant where tags are regressed on a continuous score (3/2/1/0) and thresholded on dev. The tuned ordinal variant improves macro-F1 over plain classification (0.659 vs. 0.605 on test, 5-seed means), with the gain concentrated on the SEM-DIFF minority class.

Polarity-aware calibration gate. The classifier-based SEM head can miss adversarial SEM-DIFF cases (high embedding similarity, large surface overlap). We add an independent polarity check based on word-level antonym detection: Arabic WordNet (AWN) lookup with morphological back-off, WRD-ANT hits, and lightweight rules for $\neg$ -marker proximity (لا, ليس, لم, لن) within aligned phrases. The gate fires on $19.0\% \pm 0.1\%$ of test pairs (3-seed mean); ablation results are in Section 6.4.

Figure 3 illustrates the three-layer design. Crucially, Layer 3 is *not* a downstream rescorer of Layer 1; it consumes the same word-level alignment evidence (WRD-ANT hits, negation markers) but reaches its decision through entirely rule-based logic. This factorization is deliberate: a single embedding feature distribution cannot simultaneously yield high cosine for paraphrases *and* sup-

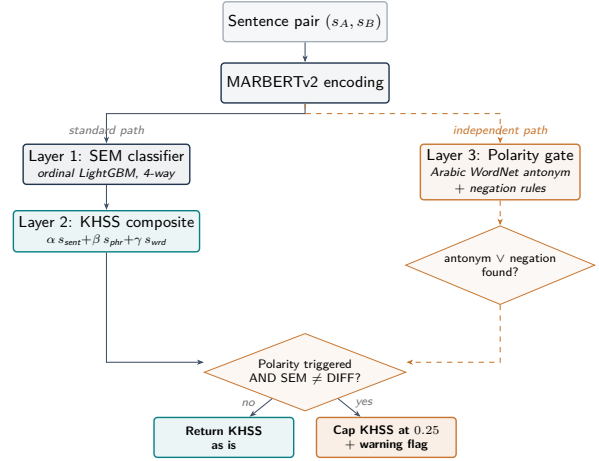


Figure 3: The polarity-aware calibration gate. Two independent paths consume the same MARBERTv2 encoding: the **standard path** (left, navy/teal) computes a similarity score KHSS; the **independent polarity gate** (right, orange) checks for antonyms or negation markers in the word-level alignment. The two paths converge at a decision step: if the polarity gate fires *and* the classifier did not already predict SEM-DIFF, KHSS is capped at 0.25 with a warning. This down-weights adversarial pairs whose surface similarity is high but polarity is reversed, a case the classifier alone tends to miss.

press them when their polarity is reversed, so the two decisions are made separately.

KHSS composite score. The KHSS combines hierarchical evidence into a single 0–1 score:

$$KHSS = \alpha s_{\text{sent}} + \beta s_{\text{phr}} + \gamma s_{\text{wrd}}, \quad (2)$$

where each level weight is modulated by a per-level confidence factor q_ℓ and the result is normalised, i.e. $KHSS = (\sum_\ell \alpha_\ell q_\ell s_\ell) / (\sum_\ell \alpha_\ell q_\ell)$ with base weights $\alpha_{\text{sent}}=0.4$, $\alpha_{\text{wrd}}=0.4$, $\alpha_{\text{phr}}=0.2$ tuned on dev, each scaled by a per-level confidence factor q_ℓ (sentence: $1 - \text{normalised entropy of the predicted SEM distribution}$; word: $\text{alignment coverage}$; phrase: $\text{count-agreement of the two predicted segmentations}$; Appendix I). s_{sent} maps SEM tags to $\{1.0, 0.66, 0.33, 0.0\}$; s_{phr} , s_{wrd} are tag-weighted averages. After computing KHSS, the polarity gate caps it at 0.25 if triggered and the predicted SEM is not already SEM-DIFF.

5 Experimental Setup

Metrics. We report macro and weighted F1 for sentence-level SEM (4-way) and phrase-level alignment (6-way). For boundary detection, we report standard BP F1 (boundary-pair) and Phrase

Span F1 (exact-span match). For the overall composite, Spearman ρ between gold SEM-ordinal and predicted KHSS. We report 5-seed (42, 7, 2024, 11, 99) means in Table 3, 3-seed (42, 7, 2024) means for ablations, and paired bootstrap (10,000 resamples) with $p < 0.05$ (Koehn, 2004).

Implementation. KASHAF is implemented in PyTorch 2.4.1 with MARBERTv2¹ loaded via Hugging Face Transformers (Wolf et al., 2020). LightGBM (Ke et al., 2017) is used for word/phrase/SEM heads, and sentence-transformers (Reimers and Gurevych, 2019) for the optional Stage 2 contrastive pre-training. Inference runs on a single CPU at ~ 50 ms/pair. We train and tune on a single NVIDIA RTX 3090. All hyperparameters are tuned on the development set; the test set is used only for final evaluation, never for model or hyperparameter selection.

Evaluation Splits. We adopt cross-batch evaluation: training data come from the 13 non-dev, non-test batches (5,809 pairs after cleaning), the development set is B13+B14 (1,000 pairs), and the test set is B03+B15 combined (1,002 pairs: $n_{\text{SEM-IDENT}} = 43$, $n_{\text{SEM-PARA}} = 198$, $n_{\text{SEM-PARTIAL}} = 718$, $n_{\text{SEM-DIFF}} = 43$). Cross-batch evaluation reduces the risk of single-batch annotator-style overfitting.

Comparison Systems. We compare against (1) majority-class and length-difference Jaccard baselines, (2) MARBERTv2 cosine (no learned head), (3) four frozen Arabic encoders with KASHAF’s ordinal head, and (4) KASHAF classification and ordinal SEM heads with all six PHR tags. Alignment-tag accuracy follows SemEval iSTS (Agirre et al., 2016) in spirit, computed over BP-matched chunks without the official phrase-to-token fan-out weighting.

LLM 3-shot prompting. We benchmark four contemporary LLMs (GPT-4o, GPT-4o-mini, Claude Sonnet 4.5, Claude Haiku 4.5) on the same cross-batch test set. We use a carefully-engineered 3-shot prompt (12 in-context examples: 3 per class) (see Appendix) that gives the schema

¹Hugging Face checkpoints: UBC-NLP/MARBERTv2 (vanilla and Stage-2 backbone), aubmindlab/bert-large-arabertv02-twitter (Antoun et al., 2020), CAMEL-Lab/bert-base-arabic-camelbert-msa (Inoue et al., 2021), and the Arabic Triplet Matryoshka v2 sentence-transformer (Nacar et al., 2025) over aubmindlab/bert-base-arabertv02.

definitions, the ordinal scale, and three labeled exemplars per class (12 in-context examples in total). The prompt is identical across all four LLMs to control for prompting effects; only the underlying model differs. Decoding temperature is fixed at 0 and outputs are parsed as JSON. We additionally evaluate all four models under zero-shot prompting and under a chain-of-thought prompt that elicits step-by-step reasoning before the label. We report all 1,002 test pairs for each model; the same metrics are computed for all systems against the same gold labels.

6 Results

6.1 Main Results

Table 3 reports the main test-set performance of KASHAF at each level of the hierarchy, alongside cosine baselines.

KASHAF achieves Spearman $\rho = 0.6715$ for sentence-level similarity (against the SEM ordinal scale), substantially above the MARBERTv2-cosine baseline ($\rho = 0.41$). On four-way SEM classification, the ordinal-regression head with tuned thresholds gives macro-F1 = 0.659 (5-seed mean, std ± 0.016 , accuracy 0.843), with the largest gains on the SEM-DIFF minority class. Phrase classification reaches macro-F1 = 0.792 over six tags with gold phrase boundaries (0.420 end-to-end); boundary detection reaches BP F1 = 0.838 and span F1 = 0.607.

6.2 LLM Baseline Comparison

Figure 5 compares KASHAF against four contemporary LLMs and the frozen-encoder baselines on the same 1,002-pair cross-batch test set; full per-class LLM scores and API costs are deferred to Table 11 (Appendix G).

KASHAF substantially outperforms all four LLMs on the four-way SEM classification task. The best LLM under 3-shot prompting (Claude Haiku 4.5) achieves macro-F1 = 0.215, more than 44 absolute points below KASHAF’s 0.659; chain-of-thought is the strongest LLM setting yet still trails by 32 points (best 0.336, Appendix G).

Few-shot vs zero-shot prompting. Re-running GPT-4o under strict zero-shot prompting (no in-context examples) yields essentially identical performance: macro-F1 0.207 vs. 3-shot 0.213 ($\Delta = -0.006$); ρ actually improves (0.526 vs. 0.505); the SEM-PARTIAL collapse is unchanged

System / Component	Macro F1	Weighted F1	Accuracy	Spearman ρ
<i>Sentence-level (SEM, 4-way classification, $n=1,002$)</i>				
Majority baseline (always SEM-PARTIAL)	0.209	0.598	0.717	—
Length-Jaccard baseline	0.270	0.480	0.580	0.34
MARBERTv2 cosine baseline	0.310	0.530	0.610	0.41
AraBERT Twitter Large (ordinal head)	0.593	0.828	0.838	—
Arabic Triplet Matryoshka (ordinal head)	0.618	0.830	0.837	—
CAMeLBERT-MSA (ordinal head)	0.617	0.823	0.827	—
MARBERTv2 vanilla (ordinal head)	0.644	0.831	0.834	—
KASHAF (classification head)	0.605	0.847	0.863	0.654
KASHAF (MARBERTv2 Stage 2, ordinal tuned)	0.659	0.841	0.843	0.6715
<i>Word alignment ($n_{\text{gold}}=4,161$ alignments)</i>				
SimAlign inter	—	—	P.709/R.944	F1 = 0.810
AwesomeAlign	—	—	P.687/R.949	F1 = 0.797
<i>Word-level tagging (5 WRD relation types; macro over 4, WRD-ANT $n=2$ excluded)</i>				
KASHAF word-level E2E (joint F1 0.828 , ALI F1 0.903)	0.590	—	0.917	—
<i>Phrase-level (6-way alignment classification)</i>				
KASHAF with <i>gold</i> phrase boundaries	0.792	0.937	0.886	—
KASHAF end-to-end (predicted everything)	0.420	0.438	—	—
<i>Phrase boundary detection (BIO segmentation)</i>				
KASHAF (CRF + MARBERTv2 layers 6–12)	—	—	BP F1 = 0.838	Span F1 = 0.607

Table 3: Main KASHAF results on cross-batch test ($n=1,002$). Sentence-level rows are 5-seed means (headline std ± 0.016); the four alternative-encoder rows share KASHAF’s ordinal-tuned head, isolating the encoder (Section 8). End-to-end PHR macro-F1 (0.420) is dominated by span-F1 (0.607); word-level E2E joint F1 is iSTS-style (Agirre et al., 2016) (correct only if both alignment and WRD tag are), computed over word pairs without the official evaluator’s phrase-to-token fan-out weighting, so not directly comparable to SemEval-2016 submissions. KASHAF’s ALI F1 (0.903, word row) is alignment accuracy on the gold-annotated word pairs, distinct from the all-token alignment F1 of the SimAlign/AwesomeAlign rows. Per-tag detail in Appendices A, B.

Predicted by GPT-4o

	I	P	Pt	D
I	39 90.7%	2 4.7%	1 2.3%	1 2.3%
P	94 47.5%	90 45.5%	9 4.5%	5 2.5%
Pt	21 2.9%	638 88.9%	27 3.8%	32 4.5%
D	1 2.3%	32 74.4%	1 2.3%	9 20.9%

88.9% of
PARTIAL
→ PARA

Figure 4: GPT-4o confusion matrix: counts and row percentages (navy shading); teal borders mark correct predictions. Gold SEM-PARTIAL (Pt) collapses overwhelmingly into the adjacent SEM-PARA (P): 638/718 (89%). The same pattern occurs across all four LLMs (I=IDENT, P=PARA, Pt=PARTIAL, D=DIFF).

(87.9% vs. 88.9%). In-context examples do not address the miscalibration in our setup.

The SEM-PARTIAL collapse. A notable pattern is that *none* of the four LLMs handle the SEM-PARTIAL class (per-class F1 in Figure 6, Appendix A). SEM-PARTIAL is the dominant gold class (71.7% of test pairs), yet all

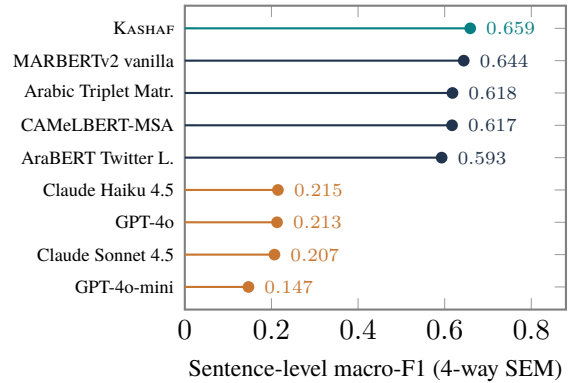


Figure 5: Sentence-level macro-F1 on the cross-batch test set ($n=1,002$). KASHAF (teal) and frozen Arabic encoder baselines (navy) share the ordinal head; LLMs (orange) use 3-shot prompting (12 examples). KASHAF reaches 0.659 versus the best 3-shot LLM’s 0.215 (+44.4 points). The best CoT result is 0.336, narrowing the gap to 32.3 points (Appendix G).

four LLMs achieve SEM-PARTIAL recall $\leq 7\%$ under zero-shot and 3-shot prompting; it is only partially alleviated under chain-of-thought (best SEM-PARTIAL F1 0.47; Appendix G). LLMs collapse SEM-PARTIAL into the adjacent SEM-PARA category (Figure 4): GPT-4o predicts SEM-PARA

for 76% of pairs (gold prior = 19.8%). LLMs reach $\rho \in [0.42, 0.53]$: coarse but not fine-grained. A trivial *majority-class* baseline (always predict SEM-PARTIAL) reaches macro-F1 0.209 (Table 3), matching the LLMs under standard prompting despite no modeling; KASHAF’s $\rho = 0.6715$ exceeds the best LLM by 0.14–0.25 points.

Cost. KASHAF runs on CPU at ~ 50 ms/pair (zero *marginal* cost; training is a one-time ~ 25 min on a single RTX 3090). GPT-4o costs \$1.50 per 1,002-pair evaluation, Claude Sonnet \$2.55.

6.3 Stage 2 Ablation

To address the question, “*is the contrastive pre-training (Stage 2) step actually contributing?*”, we re-train the phrase classifier, ordinal SEM head, and boundary detector from *vanilla* MARBERTv2 (no contrastive pre-training) across three random seeds and compare.

The pattern is nuanced. Phrase classification and untuned 4-way SEM show **no significant difference** between vanilla MARBERTv2 and Stage 2 ($p \in \{0.40, 0.80, 0.89\}$). At the three seeds shared with the vanilla baseline, the ordinal-tuned SEM head shows a small *marginal* Stage-2 advantage: 0.6530 vs. vanilla 0.6446 ($\Delta = +0.0084$, $p = 0.07$, 3-seed paired t -test; the 5-seed headline in Table 3 is 0.659); BP F1 and Span F1 show similar marginal gains ($\Delta \approx +0.005$, $p \in [0.05, 0.10]$; Table 13). All absolute differences are below 1% macro-F1: the hierarchical pipeline and feature engineering carry most of the performance, and KASHAF can be deployed without Stage 2 at a cost of only ~ 0.008 macro-F1.

6.4 KHSS Composite and Polarity-Gate Analysis

Two SEM heads, two purposes. KASHAF uses two distinct LightGBM SEM heads: the *headline* ordinal-tuned head (Table 3, $\rho = 0.6715$, 5-seed macro-F1 = 0.659) optimised for class prediction, and a separate *KHSS head* (3-seed macro-F1 = 0.589 on the same test) used solely for probability calibration in the composite below. All composite numbers use the KHSS head; ρ agrees across heads to ± 0.005 .

Component decomposition (3 seeds). In isolation against gold SEM-ordinal, $\rho(s_{\text{sent}}) = 0.660$, $\rho(s_{\text{phr}}) = 0.412$, $\rho(s_{\text{wrd}}) = 0.377$; the phrase/word terms are weaker because they use *predicted* structure (span-F1 = 0.607). Combining all three under

dev-tuned base weights (0.4, 0.4, 0.2) (confidence-scaled; Section 4) gives $\rho = 0.6484$, *slightly below* s_{sent} alone (-0.012). We therefore treat s_{phr} and s_{wrd} as *interpretive* signal localising where sentences agree or diverge (Table 5), not as scalar-correlation lift.

Polarity gate (3 seeds). The gate (Figure 3) fires on $19.0\% \pm 0.1\%$ of test pairs, raising the seed-mean ρ from 0.6484 to 0.6715 ($\Delta\rho = +0.0231$); a paired bootstrap over the 1,002 test pairs (10k resamples) gives a 95% CI of $[-0.007, +0.053]$ (one-sided $p = 0.072$), so this aggregate gain is not significant on this split. The gate is an empirical score-capping rule rather than a validated contradiction detector: it fires mostly on SEM-PARTIAL pairs and has low precision as a strict SEM-DIFF detector (0.115; Appendix E). It caps flagged high-similarity pairs to a lower ordinal target rather than re-labelling them, and since it never changes the predicted SEM class, its effect is confined to the continuous KHSS score.

7 Analysis

A worked example. Table 4 shows the full three-level analysis of a real gold pair from the test set (B03_1045). The two sentences أهلاً بك في منزلك الجديد (“welcome to your new home”, A) and مرحباً بك في منزلك الجديد (B) receive the sentence-level tag SEM-PARA: they carry the same meaning with one lexical substitution. Segmentation yields two phrases per sentence; the aligner pairs them one-to-one. The first phrase pair, أهلاً بك / مرحباً بك, is tagged PHR-SYN (the greetings أهلاً and مرحباً are synonyms), while the second, في منزلك الجديد, is tagged PHR-MATCH. At the word level, phrase 1 yields WRD-SYN for أهلاً $\leftrightarrow$ مرحباً and WRD-MATCH for بك; phrase 2 yields WRD-MATCH for all three of its words (في, الجديد, منزلك). The single lexical alternation is thus localised precisely, from sentence to phrase to the individual word, which is exactly the kind of layered trace the interpretable levels are meant to expose.

On the usability of interpretive traces. The founding motivation for iSTS is not intrinsic accuracy alone but *usability*: a system should be able to explain, to a human, why two sentences are judged similar or different. Lopez-Gazpio et al. (2017) made this explicit for English, pairing typed chunk alignments with a verbalization

#	Phrase in A	Phrase in B	Tag
<i>Phrase level</i>			
1	أهلاً بك	مرحباً بك	PHR-SYN
2	في منزلك الجديد	في منزلك الجديد	PHR-MATCH
<i>Word level (within phrase 1)</i>			
1	أهلاً	مرحباً	WRD-SYN
2	بك	بك	WRD-MATCH

Table 4: Three-level trace for pair B03_1045 (sentence-level: SEM-PARA). The synonym substitution أهلاً→مرحباً is localised to phrase 1 and, within it, to a single WRD-SYN word pair; everything else is an exact match. Word-level rows are the schema applied to the gold alignment.

module and two user studies that showed the explanations helped users. Our human evaluation (Appendix J) extends this line to Arabic for the first time: rather than measuring only agreement with gold traces (joint-F1), two linguists rated the *system-produced* phrase- and word-level traces for correctness and helpfulness on a balanced 100-pair sample. The traces are judged useful (helpfulness 3.15/5) though below the gold upper bound (−1.1), and, consistent with the intrinsic scores, helpfulness slightly exceeds correctness, i.e. a trace can help a reader localise the relation even when it is not perfectly correct. This situates KASHAF’s interpretive layers within the original usability goal of iSTS, and quantifies, for Arabic, how far automatically-produced traces still sit from human-quality explanations.

Where does KASHAF fail? Inspecting 100 errors reveals four recurring patterns. **Schema boundary cases (38%)** are pairs on the SEM-PARA/SEM-PARTIAL frontier where the annotators themselves disagreed before adjudication, an inherent ambiguity of the scheme rather than a modelling error (Appendix C). **Morphological complications (22%)** arise from Arabic’s rich inflection and clitics; the particle ما, for instance, serves four distinct grammatical functions. **Long-range coreference (15%)** degrades alignment on longer inputs, with word-level F1 dropping from 0.91 to 0.78 above 20 tokens. The remaining **25%** are stylistic and idiomatic alternations. The per-tag phrase-level breakdown, including the under-predicted REORDER tail, is in Appendix A. REORDER is in fact our hardest tag (F1 = 0.018): beyond its rarity (1.7% of gold), it has a linguistic root. In Arabic, reordering constituents

often co-occurs with a subtle semantic shift, so the sentence-level label becomes a borderline decision among SEM-IDENT, SEM-PARA, and SEM-PARTIAL, the very ambiguity our annotators faced before adjudication. This coupling of rarity with boundary ambiguity explains the low recovery.

8 Discussion

Why do LLMs collapse SEM-PARTIAL? Three signals are consistent with a calibration explanation: (i) single-label prompts bias models toward prototypes; (ii) Haiku $\approx$ Sonnet (~ 0.21 , scale invariance); (iii) SEM-PARTIAL maps to SEM-PARA but not vice versa. The collapse is most acute for the three larger models.

Why does the MARBERTv2 family win, and Stage 2 so little? KASHAF’s lead over AraBERT Twitter Large (Holm $p=0.003$) and CAMELBERT-MSA ($p=0.046$) is significant under five-seed correlation-preserving randomization; the Triplet gap is not ($p=0.108$; Appendix H). MARBERTv2’s joint MSA+dialectal pre-training matches our mostly-MSA, multi-genre register; the alternatives target Twitter, drop dialect, or optimise a coarse embedding objective. Stage 2 contrastive tuning adds only $\Delta=+0.008$ (n.s.; Section 6.3): the encoder, not auxiliary tuning, drives performance.

9 Conclusion

We presented KASHAF, the first interpretable STS resource and end-to-end system for Arabic: a 7,811-pair human-annotated corpus ($\kappa=0.83$) with a three-level schema and a modular pipeline yielding a composite KHSS score. KASHAF reaches sentence-level macro-F1 0.659 and Spearman $\rho=0.6715$; an ablation isolates the MARBERTv2 encoder, not contrastive tuning, as the driver, and a polarity-aware gate caps scores for high-similarity contradictions ($\Delta\rho=+0.0231$, n.s.). The interpretable layers localise agreement and divergence: phrase-alignment classification reaches macro-F1 0.420 and word-level tagging an iSTS-style joint F1 of 0.828. LLMs trail by 32–44 points, over-predicting SEM-PARA; typed alignment remains open. Guidelines, code, models, and a corpus sample are released; dialectal coverage is future work.

Limitations

We identify five primary limitations of this work.

First, the SEM-PARTIAL class is dominant in our test data (71.7%), reflecting the corpus’s natural distribution but creating a heavily imbalanced classification problem. While KASHAF computes macro-F1 across all four classes (so PARTIAL dominance does not inflate the headline number), the absolute class frequencies are unlike many English iSTS corpora and may overstate KASHAF’s advantage on PARTIAL-heavy data. We do not have an independent balanced Arabic test set on which to verify generalization; building one with double-annotated boundary cases between SEM-PARA and SEM-PARTIAL is a natural next step. Relatedly, our human evaluation of interpretive traces (Appendix J) shows the predicted traces are useful but rated ~ 1.1 points below gold on a 5-point scale; closing this human-level trace gap remains open.

Second, the corpus is Modern Standard Arabic with limited dialectal coverage; downstream performance on Egyptian, Levantine, or Gulf dialects is unverified. We have not systematically evaluated cross-dialectal transfer and expect a measurable drop for non-MSA text.

Third, the polarity-aware calibration gate caps the continuous KHSS score rather than re-labelling the predicted SEM class, so its contribution to SEM-DIFF classification F1 is zero because the predicted SEM label is unchanged; a gate variant that also re-classifies is a natural extension. As noted in Section 6.4, the hierarchical composite *without* the gate is slightly below s_{sent} alone (-0.012 Spearman), so we treat the phrase- and word-level terms as an interpretive overlay (Table 5) rather than a predictive lift.

Fourth, we evaluate four LLMs under zero-shot, 3-shot, and chain-of-thought (CoT) prompting (Appendix G). CoT helps (the best LLM rises from 0.215 under standard prompting to 0.336 with GPT-4o-mini) but still trails KASHAF by 32 macro-F1 points, and 60–82% of gold SEM-PARTIAL is routed to SEM-PARA even under CoT. We did not explore multi-template ensembles or task fine-tuning; a determined fine-tuned LLM could narrow the gap further, though the persistence of the dominant-class collapse across all three prompting strategies suggests the failure mode is tied to single-label ordinal prompting rather than a transient artifact. Although our corpus spans several genres (subtitles, product reviews, encyclopaedic

Wikipedia text, and scientific Arabic), it is not stratified by genre, so per-genre generalisation is not separately quantified, and coverage of legal and heavily dialectal Arabic remains limited.

Fifth, statistical power for the paired baseline comparisons is limited by the five seeds per system and the 1,002-pair test set. Correlation-preserving per-Pair_ID randomization shows the AraBERT and CAMELBERT comparisons surviving Holm–Bonferroni correction (adjusted $p=0.003$ and 0.046), whereas the Triplet Matryoshka and vanilla MARBERTv2 comparisons do not (adjusted $p=0.108$ and 0.183), even though effect sizes are large throughout ($d=0.6\text{--}3.4$; full statistics in Appendix H). We therefore frame KASHAF’s edge over the closest baselines as a *consistent cross-seed ranking* rather than a correction-significant gap, and recommend larger baseline seed counts in future work. Finally, the SEM-DIFF class, tied with SEM-IDENT as the smallest at 4.3% of the test set, carries the most annotation uncertainty: it has the lowest and widest-interval per-class F1 (0.480, 95% CI [0.353, 0.593]), and manual re-inspection indicates a subset of gold SEM-DIFF pairs are borderline with SEM-PARA. Targeted re-annotation of this class is future work and may raise measured SEM-DIFF performance.

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A Per-Tag Phrase Performance

Table 5 breaks down end-to-end phrase performance by the six PHR alignment tags, and Figure 6 shows per-class sentence-level F1 for KASHAF against GPT-4o.

PHR tag	P	R	F1	n_{gold}
PHR-MATCH	0.811	0.732	0.770	1,374
PHR-ADD_IN_B	0.448	0.550	0.494	1,243
PHR-DEL_IN_B	0.435	0.515	0.472	1,219
PHR-MOD	0.341	0.508	0.408	1,601
PHR-SYN	0.343	0.374	0.358	5,516
REORDER	0.071	0.010	0.018	192
Macro			0.420	
Micro	0.411	0.466	0.437	11,145

Table 5: Per-tag PHR results for the *end-to-end* pipeline (predicted boundaries, predicted alignments) on the cross-batch test (seed 42; other seeds within ± 0.01). PHR-MATCH (lexical overlap) is well-recovered (0.77 F1) because predicted boundaries usually capture full chunks; PHR-MOD and PHR-SYN are harder because they require fine-grained semantic distinction inside small spans. REORDER (1.7% of gold, $n=192$) is the schema’s long-tail class and severely under-predicted ($F1 = 0.018$): manual inspection of all 192 gold cases shows most involve short ambiguous chunks (≤ 4 words), a schema-design tension worth re-examining.

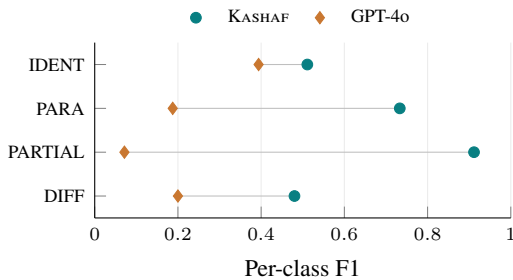


Figure 6: Per-class F1 on the cross-batch test set: KASHAF (5-seed mean, ordinal-tuned) versus GPT-4o (3-shot, from its confusion matrix, Figure 4). The gap is widest on the majority class SEM-PARTIAL (71.7% of data): KASHAF 0.91 vs. GPT-4o 0.07. On SEM-DIFF, KASHAF reaches 0.48 vs. 0.20. Matched 95% bootstrap CIs (5-seed): SEM-IDENT [0.35, 0.65], SEM-PARA [0.70, 0.77], SEM-PARTIAL [0.90, 0.93], SEM-DIFF [0.35, 0.59]; each contains its point estimate.

B Per-Tag Word Performance and iSTS Joint Correctness

Table 6 reports word-level end-to-end performance per WRD tag, together with the SemEval-iSTS-style joint correctness.

WRD tag	P	R	F1	n_{gold}
WRD-MATCH	0.996	0.981	0.989	2,919
WRD-MORPH	0.509	0.799	0.621	387
WRD-FUNC	0.566	0.421	0.484	500
WRD-SYN	0.258	0.275	0.267	353
WRD-ANT [†]	0.000	0.000	—	2
Macro (excl. ANT)			0.590	
Joint E2E F1	0.819	0.836	0.828	4,161

Table 6: Per-tag WRD end-to-end performance (3-seed mean) on the cross-batch test set. The pipeline scores each word-pair correct only when both the alignment (ALI) and the tag type (TYPE) are correct, i.e. the iSTS-style joint correctness metric. This table reports the five WRD relation types; words with no counterpart (ADD_IN_B/DEL_IN_B) are recovered by the alignment stage (ALI F1 0.903) rather than type-classified. The $n=4,161$ word decisions are the aligned word pairs across all test sentences, distinct from the 11,145 phrase-level decisions (Table 5), which additionally count unilateral ADD/DEL phrases. Performance is dominated by WRD-MATCH ($n=2,919$, 70% of gold), the easiest case. The harder distinctions are WRD-MORPH (0.621, morphological inflection) and WRD-FUNC (0.484, function-word substitution), each requiring fine-grained word-internal discrimination. WRD-SYN (0.267) is the hardest open-set class because Arabic synonymy spans a much larger lexical space than the model has direct evidence for. [†]WRD-ANT has only $n=2$ gold instances in test, both missed, giving $F1 = 0$ by the usual convention; we report the 4-class macro excluding it as an auxiliary measure given this extreme sparsity, and maintain the tag in the schema because the polarity-gate module already detects antonymy at the sentence level (Section 4, Appendix E). A future balanced challenge set (Limitations §1) will allow a robust WRD-ANT evaluation.

C Schema Decision Rules (IDENT vs. PARA vs. PARTIAL vs. DIFF)

Table 7 lists the annotator-facing decision rules that distinguish the four SEM classes, with canonical Arabic examples.

D Correspondence to SemEval iSTS

Table 8 maps our PHR and WRD tag sets onto the SemEval-2016 iSTS alignment types.

SEM tag	Decision rule (annotator-facing)	Canonical Arabic example
SEM-IDENT	Surface forms differ only in punctuation, whitespace, orthographic variants (<i>alif</i> forms, <i>ta marbuta/ha</i>), or short-vowel diacritics. No lexical, morphological, or word-order change.	A: ذهب الولد إلى السوق. B: ذهب الولد إلى السوق (no period)
SEM-PARA	All information units in A appear in B and vice versa (bidirectional entailment), but at least one differs in surface lexicalization (synonym), inflection (gender/number), or word order. No information unit is added, removed, or made more/less specific; otherwise the pair is SEM-PARTIAL.	A: هذا قرار ممتاز B: هذا قرار رائع (pure synonym: “excellent” → “great”)
SEM-PARTIAL	A and B share the topic but at least one of: (i) one side adds an information unit absent on the other; (ii) a modifier differs in specificity (e.g., temporal/quantifier); (iii) different participants/objects/locations are referenced.	A: تطبيق الخدمات المصرفية عبر الهاتف المحمول آمن وله ميزات مفيدة B: تطبيق البنك آمن وفيه خدمات مفيدة (A adds “via mobile”)
SEM-DIFF	A and B describe different events, propositions, or facts. Polarity reversal (negation/antonym) of a core predicate also forces SEM-DIFF.	A: أحب القراءة كثيرا B: أكره القراءة كثيرا (antonym of core predicate)

Table 7: SEM-tag decision rules used by annotators. Boundary cases between SEM-PARA and SEM-PARTIAL accounted for 38% of our error analysis (Section 7); the rules above formalize the most frequent disagreements observed in pilot batches. Full annotation guidelines (English + Arabic, including word-level and phrase-level decision trees) are released with the code.

KASHAF tag	SemEval iSTS relation
<i>Sentence level (SEM)</i>	
SEM-IDENT	EQUI
SEM-PARA	EQUI / SIMI
SEM-PARTIAL	SIMI / REL
SEM-DIFF	OPPO / NOALI
<i>Phrase level (PHR), chunk alignment</i>	
PHR-MATCH	EQUI
PHR-MOD	EQUI
PHR-SYN	EQUI / SIMI
PHR-REORDER	EQUI (position only)
PHR-ADD_IN_B	SPE2 / NOALI
PHR-DEL_IN_B	SPE1 / NOALI
<i>Word level (WRD)</i>	
WRD-MATCH/MORPH/FUNC	EQUI
WRD-SYN	EQUI / SIMI
WRD-ANT	OPPO
WRD-ADD_IN_B	SPE2 / NOALI
WRD-DEL_IN_B	SPE1 / NOALI

Table 8: Approximate correspondence between KASHAF tags and SemEval-2016 Task 2 iSTS chunk-alignment relations (Agirre et al., 2016). These are indicative relation mappings, not one-to-one equivalences; KASHAF tag semantics are defined in Table 1. Our schema *refines* iSTS in three ways: (i) it separates surface-preserving from surface-changing equivalence (MATCH vs. MOD/SYN/MORPH); (ii) it marks word order explicitly via REORDER, which co-occurs with EQUI; and (iii) its directional ADD_IN_B/DEL_IN_B record *which* sentence carries extra material, paralleling iSTS SPE1/SPE2. We omit the numeric iSTS similarity scores, which apply to the original chunk-alignment scale and not to KASHAF outputs.

E Polarity-Gate Behaviour by Class

Table 9 reports how often the polarity gate fires for each gold class, and its precision and recall as a SEM-DIFF detector.

F LLM Prompt Template

Table 10 gives the (abbreviated) prompt template shared by all four LLMs across the prompting conditions.

G LLM Comparison Detail

Table 11 reports the full zero-shot, 3-shot, and chain-of-thought results for all four LLMs.

H Statistical Methods and Per-Pair Significance

We quantify the significance of KASHAF’s macro-F1 advantage over each encoder baseline with a *correlation-preserving* approximate-randomization test over all 5 seeds (10,000 iterations). Because the same test pair recurs across seeds, we do not pool the $5 \times 1,002$ predictions as independent observations; instead, for each Pair_ID we draw a single swap decision and apply it jointly across all seeds, then recompute the seed-averaged macro-F1 difference, avoiding pseudo-replication. Table 12 reports the macro-F1 difference Δ , the two-sided randomization

Gold class	n	Fires	Rate
SEM-IDENT	43	0	0.0%
SEM-PARA	198	3	1.5%
SEM-PARTIAL	718	166	23.1%
SEM-DIFF	43	22	51.2%
Total	1,002	191	19.1%
Gate as SEM-DIFF detector: $P = 0.115$, $R = 0.512$, $F1 = 0.188$			
Aggregate Spearman: $\Delta\rho = +0.0231$, bootstrap 95% CI $[-0.007, +0.053]$ (n.s.)			

Table 9: Polarity gate behaviour by gold class (seed 42; other seeds within ± 1 pair). The gate never fires on SEM-IDENT and rarely on SEM-PARA (1.5% of 198 pairs), recovers 51% of SEM-DIFF gold, but is most active on SEM-PARTIAL (23%); over all non-SEM-DIFF classes its false-positive rate is $169/959 \approx 17.6\%$. Its low precision as a strict SEM-DIFF detector (0.115) reflects that SEM-PARTIAL pairs with detected antonym/negation cues (e.g. partial polarity reversal) are score-capped to 0.25, which aligns the score with their lower ordinal target. The resulting aggregate Spearman gain ($\Delta\rho = +0.0231$) is positive but not statistically significant on the test split (paired-bootstrap 95% CI $[-0.007, +0.053]$, one-sided $p = 0.072$); the gate is, by design, an empirical score-capping rule rather than a class re-labeler, and we do not claim it as a validated contradiction detector.

Prompt structure (identical across all four LLMs; 12 in-context examples for 3-shot, 0 for zero-shot)

System. You are an Arabic Semantic Textual Similarity annotator. You will be given two Arabic sentences and you must classify their semantic relation using exactly one of four tags: SEM-IDENT, SEM-PARA, SEM-PARTIAL, SEM-DIFF.

Tag definitions.

- SEM-IDENT: identical content; differ only in punctuation, whitespace, or short-vowel diacritics.
- SEM-PARA: paraphrases; same propositional content with different surface form (synonyms, inflection, word order).
- SEM-PARTIAL: share topic but at least one information unit, modifier specificity, or participant differs.
- SEM-DIFF: describe different events/facts, or one sentence reverses polarity of a core predicate.

3-shot examples (12 total; 3 per class, randomly sampled once from the training set, fixed across all 4 LLMs):

Example 1. A: [Arabic sentence]; B: [Arabic sentence]; Label: SEM-IDENT

Example 2. A: [Arabic sentence]; B: [Arabic sentence]; Label: SEM-IDENT

... (3 examples per class for all four classes) ...

Task. A: [Arabic sentence A]

B: [Arabic sentence B]

Output format. Return a JSON object: {"label": "<one of the four tags>", "reasoning": "<one sentence in Arabic>"}. Do not return anything else.

Table 10: LLM prompt template (translated/abbreviated for space; full prompt is released with the code). For zero-shot, the 12 in-context examples are omitted; all other instructions remain identical. Decoding temperature = 0; max tokens = 80.

p -value, Cohen’s d over seed-level means, and the Holm–Bonferroni-adjusted p -value over the four-comparison family (FWER=0.05). After correction, the AraBERT and CAMELBERT comparisons are significant, while Triplet and MARBERTv2-vanilla are not, consistent with the marginal Stage-2 effect in Table 13.

Pairwise significance claims are based on the correlation-preserving approximate-randomization test reported in Table 12. The observed macro-F1 differences are positive for all four comparisons, but only the AraBERT and CAMELBERT comparisons remain significant after Holm–Bonferroni correction.

I Reproducibility: Hyperparameters and Feature Sets

This appendix specifies every hyperparameter, the dev-tuning choices, and the exact feature sets, so that each reported number can be traced through the released code. Unless noted, defaults below are the values used for all reported runs.

Encoder and embeddings. The backbone is MARBERTv2 (Abdul-Mageed et al., 2021) (163M parameters; 12 layers; hidden size 768; 12 attention heads; intermediate size 3072; vocabulary 100,000; float32). Word- and phrase-alignment features use frozen **layer-8** token embeddings (selected on dev). The boundary detector instead

System / prompting	Macro F1	SEM-PARTIAL F1
<i>Zero-shot prompting</i>		
GPT-4o-mini	0.147	0.136
GPT-4o	0.207	0.086
Claude Haiku 4.5	0.162	0.099
Claude Sonnet 4.5	0.156	0.079
<i>3-shot prompting (12 in-context examples)</i>		
GPT-4o-mini	0.147	0.119
GPT-4o	0.213	0.071
Claude Haiku 4.5	0.215	0.086
Claude Sonnet 4.5	0.207	0.079
<i>Chain-of-thought prompting (zero-shot)</i>		
GPT-4o-mini	0.336	0.467
GPT-4o	0.303	0.340
Claude Haiku 4.5	0.291	0.250
Claude Sonnet 4.5	0.324	0.298
KASHAF (ordinal, tuned)	0.659	0.912

Table 11: LLM baselines across three prompting strategies versus KASHAF on the cross-batch test set ($n=1,002$). Model IDs: gpt-4o-2024-08-06, gpt-4o-mini-2024-07-18, claude-sonnet-4-5-20250929, claude-haiku-4-5; decoding temperature 0. Across *all twelve* LLM configurations, 60–91% of gold SEM-PARTIAL is routed to SEM-PARA (the dominant-class collapse), versus KASHAF’s SEM-PARTIAL recall of 0.947. Chain-of-thought is the strongest LLM setting (best 0.336, GPT-4o-mini) yet still trails KASHAF by 32 macro-F1 points; under zero-shot and 3-shot, SEM-PARTIAL recall is $\leq 7\%$. 3-shot lifts macro-F1 modestly over zero-shot (Haiku 0.162 $\rightarrow$ 0.215) yet leaves this collapse intact. KASHAF’s training is a one-time ~ 25 min on a single RTX 3090.

uses the per-token mean of **layers 6–12** of a fine-tuned encoder. Inputs are Arabic-normalised (diacritic removal, alif/ya unification) before tokenisation. Maximum sequence length is 128 tokens for the SEM head and 64 for the boundary detector.

LightGBM tag classifiers (SEM, phrase, word).

All three heads share the same core configuration, tuned on dev: objective multiclass with multi_logloss; learning rate 0.05; num_leaves = 31; min_data_in_leaf = 20; feature_fraction = 0.9; bagging_fraction = 0.9 with bagging_freq = 5; L1/L2 regularisation $\lambda_1=\lambda_2=0.1$; and 500 boosting rounds. Class imbalance is handled with per-class sample weights.

Ordinal SEM head. The headline SEM model is a LightGBM *regression* on the ordinal scalar target $\{0, 1, 2, 3\}$ (SEM-DIFF=0 to SEM-IDENT=3). Continuous predictions are converted to classes with three thresholds initialised at $[0.5, 1.5, 2.5]$

and then tuned on dev by maximising macro-F1 (Nelder–Mead). Tuned thresholds improve macro-F1 over the default thresholds, with the gain concentrated on the minority SEM-DIFF class. We report 5-seed means (seeds 42, 7, 2024, 11, 99).

Boundary detector (BIO + CRF). Fine-tuned MARBERTv2 with a linear layer and a CRF, trained for 5 epochs; batch size 32; learning rate 2×10^{-5} ; Adam (Kingma and Ba, 2015); maximum length 64.

Word alignment. SimAlign (Jalili Sabet et al., 2020) with the inter matching method over layer-8 embeddings, producing the sparse alignment set $\mathcal{A}_w$ consumed by the phrase- and word-level classifiers.

KHSS composite. Base level weights (tuned on dev) are $\alpha_{\text{sent}}=0.4$, $\alpha_{\text{word}}=0.4$, $\alpha_{\text{phr}}=0.2$. Each weight is scaled by a per-level confidence factor q_ℓ , and the weighted sum is normalised by $\sum_\ell \alpha_\ell q_\ell$ (Section 4). The confidence factors use only predicted quantities (no gold): $q_{\text{sent}}=1-\hat{H}$, where $\hat{H}$ is the normalised entropy of the predicted SEM probability vector (high when the classifier is confident); q_{word} is word-alignment coverage, the fraction of source tokens receiving an alignment ($|\text{aligned}|/|\text{tokens}|$); and $q_{\text{phrase}}=\min(n_p, \hat{n}_p)/\max(n_p, \hat{n}_p)$ is a count-agreement between the two predicted segmentations (not boundary accuracy). Each returns a value in $[0, 1]$. Because every sentence contains at least one token and at least one phrase, $q_{\text{phrase}}>0$, so the normalising denominator $\sum_\ell \alpha_\ell q_\ell$ is never zero; the fallback $q=0$ for an empty per-level denominator is therefore never reached in practice. Within phrase pairing, alignment and semantic evidence are combined as $\text{sim} = \alpha_{\text{pair}} \cdot \text{align} + (1 - \alpha_{\text{pair}}) \cdot \text{sem}$ with $\alpha_{\text{pair}}=0.5$; this α_{pair} is distinct from the KHSS level weights above. The phrase score s_{phr} uses a per-tag value table (e.g., REORDER=0.30; unmatched default 0.5). If the polarity gate fires and the predicted SEM is not already SEM-DIFF, KHSS is capped at 0.25.

Feature sets. The word-level classifier uses ≈ 22 features and the phrase-level classifier uses a distinct ≈ 23 -feature set; both combine surface and semantic groups. *Surface features* include exact match, token- and character-length statistics and ratios, Levenshtein distance (raw and normalised), unigram/bigram/trigram Jaccard, prefix/suffix overlap, stem match, and directional

Comparison	Δ	raw p	Cohen’s d	Holm p
KASHAF vs AraBERT Twitter L.	+0.066	0.0007	+3.4	0.003
KASHAF vs CAMELBERT-MSA	+0.042	0.015	+2.2	0.046
KASHAF vs Triplet Matryoshka	+0.041	0.054	+2.0	0.108
KASHAF vs MARBERTv2 vanilla	+0.015	0.183	+0.6	0.183

Table 12: Correlation-preserving approximate-randomization significance of KASHAF vs each baseline (all 5 seeds; swap decisions drawn per Pair_ID and applied jointly across seeds; 10^4 iterations). Holm–Bonferroni over $m=4$ comparisons: the AraBERT and CAMELBERT comparisons survive FWER=0.05. Cohen’s d is computed over seed-level macro-F1 means.

Component / Metric	Stage 2	Vanilla	Δ (V–S2)	p -value
Phrase classifier (6-tag, 3-seed mean)				
Macro-F1	0.7927	0.7921	−0.0006	0.80
Weighted-F1	0.9366	0.9363	−0.0003	0.89
Ordinal SEM (4-tag, 3-seed mean, tuned thresholds)				
Classification Macro-F1	0.5996	0.5929	−0.0067	0.40
Ordinal Tuned Macro-F1	0.6530	0.6446	−0.0084	0.07
BoundaryDetector (CRF) (3-seed mean)				
BP F1	0.8411	0.8375	−0.0036	0.05
Phrase Span F1	0.6157	0.6075	−0.0082	0.08

Table 13: Stage 2 ablation: re-training all components with vanilla MARBERTv2 vs. Stage 2 contrastive backbone (3 seeds, paired t -test). Phrase classification and untuned SEM show no significant difference ($p \in \{0.40, 0.80, 0.89\}$); boundary metrics and tuned SEM show only marginal differences ($|\Delta| < 0.01$). The polarity-aware calibration gate ablation is reported separately in Section 6.4 ($\Delta\rho = +0.0231$, paired-bootstrap 95% CI $[-0.007, +0.053]$, not significant).

word overlap. *Semantic features* are embedding cosine similarity, L2 distance, and mean/max of the embedding difference vector. Phrase features additionally include the `a_is_none/b_is_none` indicators for unaligned phrases. The polarity gate additionally consults Arabic WordNet (AWN) antonym lookups and Farasa (Abdelali et al., 2016) morphological distance.

Environment and cost. Training uses a single NVIDIA RTX 3090 (Python 3.11, PyTorch 2.4.1+cu124, LightGBM 4.x). Inference runs on CPU at ≈ 50 ms/pair; full training is a one-time ≈ 25 minutes. All hyperparameters are tuned on dev; the test set is used only for final evaluation.

J Human Evaluation of Interpretability

Intrinsic joint-F1 measures whether predicted traces match the gold annotation, but not whether a human reader finds them *usable*. To assess this directly, two trained linguists (not involved in system development) evaluated the predicted phrase- and word-level traces on a stratified sample of 100 test pairs, balanced across the four SEM classes (≈ 25 pairs each) so that the minority classes are not drowned out by the dominant SEM-PARTIAL class.

Protocol. For each pair, the two judges independently rated the interpretive trace on two five-point scales: *correctness* (is the trace linguistically right?) and *helpfulness* (does it help a reader see why the two sentences match or differ?). Predicted and gold traces were shown *blind and unlabeled*, so a judge could not tell which one the system produced; this also yields the gold trace as a natural upper bound. Inter-annotator agreement was $\kappa = 0.65$ (substantial), consistent with our original annotation agreement (SEM $\kappa = 0.83$; phrase/word 0.79/0.77).

Results. Table 14 reports mean ratings. The predicted traces are rated *useful and mostly correct* (system correctness 3.0, helpfulness 3.15 out of 5), with a consistent gap below the gold traces (−1.1 overall on correctness; −1.2 at the phrase level and −1.0 at the word level). Helpfulness slightly exceeds correctness, indicating that a trace can help a reader localise the relation even when it is not structurally perfect. These ratings agree with the intrinsic scores (phrase/word joint-F1 in Appendices A and B): the interpretive layers provide a useful signal without yet reaching human-level trace quality. We therefore treat trace usability as a genuine but partial strength, and close the human-level gap as

an open direction (Limitations).

Level	Sys. corr.	Sys. help.	Gold corr.	Gap
Phrase	3.0	3.3	4.2	-1.2
Word	3.0	3.0	4.0	-1.0
Overall	3.0	3.2	4.1	-1.1

Table 14: Human evaluation of interpretive traces on a balanced 100-pair sample (≈ 25 per SEM class), two linguist judges, $\kappa = 0.65$. Ratings are 1–5 for system-predicted (Sys.) and gold traces, shown blind. *Gap* is system minus gold correctness.

K Ethics, Data, and Annotator Statements

K.1 Data Statement

Following [Bender and Friedman \(2018\)](#), we report key data properties.

Curation rationale. The corpus targets Arabic interpretable Semantic Textual Similarity (iSTS) at sentence, phrase, and word levels. Source sentences are drawn from several public Arabic sources: movie and television subtitles ([Lison and Tiedemann, 2016](#)), product reviews, encyclopaedic articles from Wikipedia, and scientific texts. Pairs were selected to populate four SEM tags with explicit attention to the SEM-DIFF and SEM-IDENT minority classes; curated paraphrase pools and adversarial near-paraphrase distractors were added; the resulting training distribution (SEM-IDENT 4.6%, SEM-PARA 32.3%, SEM-PARTIAL 56.4%, SEM-DIFF 6.7%) is less skewed than the natural test distribution but still SEM-PARTIAL-dominated.

Language variety. Modern Standard Arabic (MSA) is the dominant variety across sources (reviews, encyclopaedic, and scientific text are predominantly MSA; subtitles are largely MSA with occasional dialectal lexical items). The corpus is not stratified by Arabic dialect, and dialectal coverage (Egyptian, Levantine, Gulf, Maghrebi) is unverified (Limitations §2).

Speaker / annotator demographics. Source-text authors are not identifiable: subtitles are translated screen text, and user-generated review text was de-identified before annotation. Three trained annotators participated; all are adult native speakers of Arabic with university-level training in Ara-

bic linguistics, of Saudi and Egyptian backgrounds. The senior reviewer is a faculty linguist.

Text characteristics. Mean sentence length: 8.4 tokens (range 2–47). Genre: mixed, comprising conversational (subtitles), user-generated (product reviews), encyclopaedic (Wikipedia), and scientific Arabic. All source text was collected from publicly accessible sources; user-generated text was de-identified, and the released corpus contains no personally identifying information.

K.2 Annotator Compensation and Working Conditions

Annotators were compensated at a rate aligned with senior research-assistant pay scales in the host institution, well above the local minimum wage and consistent with prior Arabic-NLP annotation projects of comparable difficulty. Annotation guidelines were provided in advance (100+ page bilingual guide), with iterative feedback over multiple pilot batches before production annotation. All annotators were full participants in schema refinement and could flag disputed items for adjudication.

K.3 Ethics Statement

The corpus is assembled from publicly available text across the sources above. No human-subjects data, medical records, or sensitive personal information are involved, and user-generated material was de-identified. Each source retains its upstream license (e.g., Wikipedia under CC BY-SA; subtitles under the OpenSubtitles terms); the released subset and audit scripts are distributed for non-commercial academic research, and redistribution of any source text follows the most restrictive applicable upstream terms; access to the full corpus is governed by the same terms (Section K.4).

Potential dual-use considerations. Interpretable Semantic Textual Similarity systems can support beneficial applications (information retrieval, plagiarism detection, machine-translation quality estimation) and could in principle be used for content monitoring at scale. Our schema and system do not enable any capability not already supplied by general-purpose Arabic NLP toolkits, and we view the resource contribution (annotated benchmark, open code) as net-positive for the Arabic NLP research community.

Use of AI assistance. We used a large language model for editorial polish of the English prose; all research design, experiments, data construction, annotation, results computation, and scientific claims are the authors’ own work, and the authors take full responsibility for all content of the paper. All numerical results are computed from artefacts in our experiment runs directory and are reproducible from the released code.

K.4 License and Access

KASHAF’s guidelines, code, models, pipeline, a corpus sample, and the audit script (Section 3) are released under a non-commercial research license. The full annotated corpus is hosted by the King Salman Global Academy for the Arabic Language (KSAA) and is available to researchers on request, free of charge, for non-commercial research under a data-use agreement; distribution remains subject to the upstream terms of each source (Section K.3).